

Shadman Sakib

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RESEARCH PROFILE

Ph.D. researcher in **Information Systems (AI/ML Track)** at the University of Maryland, Baltimore County (UMBC), developing **reliable, deployable, and evidence-grounded multimodal AI for contactless physiological sensing, trustworthy medical reasoning, and autonomous health monitoring**. Research integrates computer vision, video/time-series modeling, vision-language models, signal processing, uncertainty and reliability estimation, and on-device edge inference. Work has progressed from algorithm development to **peer-reviewed publications, field demonstrations, and testbed validation**, including collaborative experimentation with the **DEVCOM Army Research Laboratory (ARL)** and NSF-supported research infrastructure at UMBC.

EDUCATION

Ph.D. in Information Systems (AI/ML Track)

Jan. 2024 – Present

University of Maryland, Baltimore County (UMBC)

Maryland, United States

- CGPA: **4.00/4.00**; invited to join The Honor Society of Phi Kappa Phi. Research emphasis: multimodal AI, computer vision, physiological sensing, trustworthy AI, medical reasoning, ubiquitous computing, and edge intelligence.

M.Sc. in Information Systems (AI/ML Track)

Jan. 2024 – Aug. 2025

University of Maryland, Baltimore County (UMBC)

Maryland, United States

- CGPA: **4.00/4.00**. Graduate work in AI for biomedical applications, medical signal processing, machine learning optimization, NLP, data mining, and statistical learning.

B.Sc. in Electrical and Electronic Engineering

Aug. 2015 – Aug. 2019

International University of Business Agriculture and Technology (IUBAT)

Dhaka, Bangladesh

- CGPA: **3.96/4.00**; Rank: **1/66 (Top 1%)**.

RESEARCH AND DEVELOPMENT EXPERIENCE

Graduate Research Assistant, Mobile, Pervasive and Sensor Computing (MPSC) Lab

Jan. 2024 – Present

University of Maryland, Baltimore County (UMBC)

Baltimore, Maryland

- **Project 1 — Reliable Contactless Physiological Sensing from RGB/NIR/IR Video**
 - Develop multimodal vision and signal-processing pipelines for respiratory-rate and physiological monitoring from subtle respiration-induced body motion, with emphasis on robustness to motion, viewpoint, lighting, occlusion, and signal-quality variation.
 - Developed **RespFormer**, a motion-guided temporal-frequency multimodal transformer for respiratory-rate regression and breathing-pattern classification; developed **E2RespUNet**, an attention-enhanced U-Net for end-to-end respiratory waveform reconstruction and rate estimation; optimized inference for NVIDIA Jetson edge hardware.
 - Contributed to **RRPIPS**, formulating respiratory waveform recovery as persistent multi-point motion tracking with coarse-to-fine association and temporal consistency; curated and synchronized multimodal datasets and implemented SQI-based ROI selection, standardized preprocessing, ablation studies, and cross-modal evaluation.
 - Extended this research toward reliability-guided video representation learning, quality-aware sensing, and deployable contactless physiological monitoring under heterogeneous sensing conditions.

Selected outputs: IEEE ICMLA 2025 • IEEE SmartComp 2025 • IEEE/ACM CHASE 2025 • IEEE/ACM CHASE 2026 • ACM CIKM 2026 (**accepted**)
- **Project 2 — Trustworthy Vision-Language and Evidence-Grounded Reasoning for Medical AI**
 - Develop reliability-aware multimodal reasoning methods that explicitly separate sensing and perception quality from downstream reasoning, calibrate available evidence, audit candidate predictions, and reduce unsupported conclusions in physiological and medical AI workflows.
 - Developed **VLS-RR**, a perception–reasoning framework for contactless respiratory monitoring; **CALIBRA**, a calibration-aware multi-agent verification framework; and **EviCal-RR**, an evidence-calibrated reasoning frame-

work for reliable physiological inference.

- Investigate active evidence acquisition, selective auditing, counterfactual auditing, grounded evidence verification, and quantitative temporal reasoning in physiological video and multimodal health agents.

Selected outputs: IEEE/ACM CHASE 2026 • CVPR 2026 GRAIL-V • MICCAI 2026 SAFER • AAAI 2026 Fall Symposium (accepted) • NeurIPS 2026 Med-Reasoner (2 papers accepted)

Current manuscripts: IEEE/CVF WACV 2027 • ACM Transactions on Computing for Healthcare

- **Project 3 — Multimodal Foundation Models and Agentic Physiological AI**

- Build video–time-series and VLM-guided models that align visual, physiological, and contextual streams for contactless monitoring, cross-subject generalization, and low-data adaptation.
- Investigate masked temporal modeling, multi-scale contrastive alignment, parameter-efficient fine-tuning (PEFT), uncertainty-aware evaluation, fairness across sensing conditions, and efficient edge deployment through quantization and knowledge distillation.
- Developed **ViTaL-Diff**, a video-token latent diffusion framework for respiratory waveform reconstruction with uncertainty-aware physiological inference.
- Explore agentic multimodal workflows in which foundation models coordinate perception, evidence assessment, physiological reasoning, verification, and reporting across visual and wearable sensing streams.

Selected outputs: ICML 2026 SD4H Workshop

Graduate Research Assistant, Center for Real-time Distributed Sensing and Autonomy (CARDS) May 2025 – Aug. 2025

UMBC – Collaborative Research with DEVCOM Army Research Laboratory (ARL)

Maryland, United States

- **Project — Autonomous Contactless Physiological Sensing on Mobile Robots using DVPG**

- Developed an **on-device contactless physiological sensing system** for autonomous respiratory-rate estimation on a Boston Dynamics Spot robot, deploying the complete computer-vision and signal-processing pipeline on NVIDIA Jetson Orin.
- Designed robust respiratory-motion extraction using ROI detection, optical flow, temporal filtering, distance profiling, camera-geometry compensation, and signal-quality assessment under changing viewpoints, outdoor illumination, partial occlusion, and robot-induced motion.
- Conducted field experimentation at the **DEVCOM Army Research Laboratory Robotics Research Collaboration Campus (R2C2), Graces Quarters, Maryland**, evaluating the system under realistic robotic operating conditions.
- Contributed to the **2025 ArtIAMAS University Experimentation Event** and Collaborative Remote Robotics Experiment using the Distributed Virtual Proving Ground (DVPG). [ArtIAMAS Field Experiment](#)
- Extended the research to heterogeneous mobile robots and multimodal edge sensing for field respiratory monitoring.

Selected outputs: IEEE DCROSS-IoT 2026.

Graduate Research Assistant, Quantitative Ultrasound Imaging (QUEST) Lab

Jan. 2023 – Dec. 2023

University of North Carolina at Charlotte

Charlotte, North Carolina

- **Project — Deep Learning and Quantitative Ultrasound for Obstetric Imaging**

- Investigated non-invasive tissue-imaging biomarkers for early disease screening using quantitative ultrasound, signal processing, deep learning, optimization, multimodal image segmentation, acquisition, and registration workflows.

Research Assistant, Advanced Medical Engineering Research Institute (AMERI)

Sep. 2019 – Sep. 2020

University of Hyogo

Hyogo, Japan

- **Project — AI-Assisted Surgical Video Understanding and Navigation**

- Developed deep-learning methods for real-time processing of wearable smart-glass surgical video, including surgical-phase recognition, hand-frame extraction, surgical-tool detection, and context-aware surgical workflow understanding.

Media feature: [AffordSENS Media Feature](#)

Selected outputs: Joint IEEE SCIS-ISIS 2020

RESEARCH DEMONSTRATIONS AND FIELD VALIDATION

- **NSF TRACE Testbed Workshop and Field Experiment Demonstration**, CYPRESS Technology Research Center, UMBC Apr. 2026

Participated in field experimentation and research demonstrations using the NSF TRACE Testbed for autonomous and cyber-physical systems research. [Demo Gallery](#) | [Workshop Page](#)

- **Collaborative Remote Robotics Experiment using the Distributed Virtual Proving Ground (DVPG)**, DEVCOM ARL R2C2, Graces Quarters, Maryland Aug. 2025
Demonstrated on-device contactless physiological sensing on a Boston Dynamics Spot robot during the ArtIAMAS University Experimentation Event. [Public Event Evidence](#)

SELECTED PUBLICATIONS

Full publication record: [Google Scholar](#).

Peer-Reviewed Journal Articles

1. **Sakib, S.**, Hasan, Z., & Roy, N. A State-Of-The-Art Survey of Remote Photoplethysmography for Contactless Health Parameters Sensing. *WIREs Data Mining and Knowledge Discovery*, 15(3), 2025. **Corresponding author**.
2. Shinde, G., Ravi, A., Dey, E., **Sakib, S.**, Rampure, M., & Roy, N. A Survey on Efficient Vision–Language Models. *WIREs Data Mining and Knowledge Discovery*, 2025.
3. Ahmed, I., Routh, B. B., Kohinoor, M. S. R., **Sakib, S.**, Rahman, M. M., & Azzedin, F. Multi-Model Attentional Fusion Ensemble for Accurate Skin Cancer Classification. *IEEE Access*, 2024.
4. Hasib, K. M., Islam, M. R., **Sakib, S.**, Akbar, M. A., Alam, M. S., & Razzak, I. Depression Detection From Social Networks Data Based on Machine Learning and Deep Learning Techniques: An Interrogative Survey. *IEEE Transactions on Computational Social Systems*, 10(4), 1568–1586, 2023.

Selected Conference and Workshop Proceedings

1. **Sakib, S.** & Roy, N. Measure What Matters: Active Evidence Acquisition for Reliable Physiological Reasoning. *NeurIPS 2026 Workshop on Medical Reasoning with Vision Language Foundation Models (Med-Reasoner)*, Atlanta, Georgia, USA, 2026. **Accepted**.
2. **Sakib, S.** & Roy, N. Can VLMs Keep Time? Probing Quantitative Temporal Reasoning in Physiological Videos. *NeurIPS 2026 Workshop on Medical Reasoning with Vision Language Foundation Models (Med-Reasoner)*, Atlanta, Georgia, USA, 2026. **Accepted**.
3. Sami, A. A., Absar, S., **Sakib, S.**, Ahmed, I., & Tushar, Z. H. Do LLM Agents Follow Therapeutic Boundaries? Evaluating Cross-System Drug Substitution Hallucination. *AAAI 2026 Fall Symposium on Agentic and Trustworthy AI for Health and the Global AI-Ready Nursing Workforce*, 2026. **Accepted**.
4. **Sakib, S.** & Roy, N. Does the Agent Follow the Evidence? Counterfactual Auditing for Trustworthy Multimodal Health Agents. *AAAI 2026 Fall Symposium on Trustworthy Agentic Systems (TAS 2026)*, Arlington, Virginia, USA, 2026. **Accepted**.
5. **Sakib, S.**, Patel, H., & Roy, N. Reliability-Guided Video Token Learning for Contactless Vital Sign Estimation. *35th ACM International Conference on Information and Knowledge Management (CIKM 2026)*, Rome, Italy, 2026. **Accepted**.
6. **Sakib, S.** & Roy, N. EviCal-RR: Evidence-Calibrated Reasoning for Contactless Respiratory Monitoring. *MICCAI 2026 Workshop on Stable Adaptation and Faithful Evaluation of Reasoning in Medical Foundation Models (SAFER)*, Strasbourg, France, 2026. **Accepted**.
7. **Sakib, S.**, Shinde, G., & Roy, N. Decoupling Perception and Reasoning for Contactless Respiratory Rate with Vision Language and Small Language Models. *IEEE/ACM CHASE 2026*, Pittsburgh, Pennsylvania, 2026.
8. **Sakib, S.**, Rampure, M., & Roy, N. A Quality-Aware Framework for Contactless Respiratory Monitoring Under Variable Sensing Conditions. *2nd Workshop on Sensing and Computing for Smart and Connected Health (SCH)*, co-located with *IEEE/ACM CHASE 2026*, Pittsburgh, Pennsylvania, 2026.
9. **Sakib, S.** & Roy, N. ViTaL-Diff: Video-Token Latent Diffusion for Contactless Respiratory Monitoring. *ICML 2026 Workshop on Structured Data for Health (SD4H)*, Seoul, South Korea, 2026.
10. Rampure, M.^{*}, **Sakib, S.**^{*}, Patel, H., Hasan, Z., & Roy, N. Contactless Respiratory Monitoring on Heterogeneous Mobile Robots: A Multimodal Edge-Computing Framework. *8th International Workshop on IoT Applications and Industry 5.0*, co-located with *IEEE DCROSS-IoT 2026*, Reykjavik, Iceland, 2026. ^{*}Equal contribution.
11. **Sakib, S.**, Shinde, G., & Roy, N. CALIBRA: Calibration-Aware Multi-Agent Verification for Contactless Physiological Monitoring. *CVPR 2026 Workshop on Grounded Retrieval and Agentic Intelligence for Vision-Language (GRAIL-V)*, Denver, Colorado, 2026.
12. **Sakib, S.**, Shinde, G., Chugh, S., Anwar, M. S., & Roy, N. RespFormer: A Motion-Guided Temporal-Frequency Multimodal Fusion Transformer for Contactless Respiratory Monitoring. *IEEE ICMLA 2025*, Boca Raton, Florida, 2025.
13. **Sakib, S.**, Shinde, G., Dey, E., & Roy, N. E2RespUNet: End-to-End Respiratory Signal Reconstruction and Rate Prediction Using a Unified Attention-Enhanced U-Net. *IEEE SmartComp 2025*, Cork, Ireland, 2025.
14. Hasan, Z., Ahmed, M., **Sakib, S.**, Chugh, S., Khan, M. A., Faridee, A. Z. M., & Roy, N. RRPIPS: Respiratory Waveform

Reconstruction using Persistent Independent Particles Tracking from Video. *ACM/IEEE CHASE 2025*, New York, United States, 2025.

15. **Sakib, S.**, Chowdhury, M. J. U., Hira, M. M. H., Choudhury, T. E., & Das, S. Deep Learning-Based Sentiment Analysis of Social Media and E-Commerce Reviews. *IEEE International Conference on Computing and Machine Intelligence (ICMI 2025)*, Michigan, United States, 2025.
16. **Sakib, S.**, Siddiky, M. N. A., Arifuzzaman, M., & Chowdhury, M. J. U. Optimizing Facial Recognition: An Analytical Comparison of Traditional and Deep Learning Approaches. *IEEE International Conference on Data Science and Its Applications (ICoDSA 2024)*, Bali, Indonesia, 2024.
17. Hasib, K. M., **Sakib, S.**, Mahmud, J. A., Mithu, K., & Rahman, M. S. COVID-19 Prediction based on Infected Cases and Deaths of Bangladesh using Deep Transfer Learning. *IEEE World AI IoT Congress (AIoT 2022)*, Seattle, United States, 2022.
18. **Sakib, S.**, Hasib, K. M., Tasawar, I. K., Tanzeem, A. K., Arefin, M. F., Islam, S., & Alam, M. S. A Data-Driven Hybrid Optimization Based Deep Network Model for Short-Term Residential Load Forecasting. *IEEE IEMCON 2021*, Vancouver, Canada, 2021. **Best Paper Award, Signal Processing Track.**
19. **Sakib, S.**, Tanzeem, A. K., Tasawar, I. K., Shorna, F., Siddique, M. A. B., & Binte Alam, S. Blood Cancer Recognition Based on Discriminant Gene Expressions: A Comparative Analysis of Optimized Machine Learning Algorithms. *IEEE IEMCON 2021*, Vancouver, Canada, 2021.
20. **Sakib, S.**, Yasmin, N., Tanzeem, A. K., Shorna, F., Hasib, K. M., & Binte Alam, S. Breast Cancer Detection and Classification: A Comparative Analysis Using Machine Learning Algorithms. *3rd International Conference on Communication, Computing and Electronics Systems (ICCCES 2021)*, Springer, 2021.
21. **Sakib, S.**, Yasmin, N., Tasawar, I. K., Aziz, A., Siddique, M. A. B., & Khan, M. M. R. Performance Analysis of Machine Learning Approaches in Diabetes Prediction. *IEEE Region 10 Humanitarian Technology Conference (R10 HTC 2021)*, Bangalore, India, 2021. **Best Paper Award, Biomedical Engineering Track.**
22. Shishir, F. S., Hasib, K. M., **Sakib, S.**, Maitra, S., & Shah, F. M. De Novo Drug Property Prediction using Graph Convolutional Neural Networks. *IEEE R10 HTC 2021*, Bangalore, India, 2021.
23. **Sakib, S.**, Siddique, M. A. B., Khan, M. M. R., Yasmin, N., Aziz, A., Chowdhury, M., & Tasawar, I. K. Transfer Learning Based Method for Automatic COVID-19 Cases Detection in Chest X-Ray Images. *IEEE International Conference on Smart Electronics and Communication (ICOSEC 2021)*, Trichy, India, 2021.
24. **Sakib, S.**, Hossain, B., Hiranaka, T., & Kobashi, S. Hand Frame Extraction in Surgical Video Images Using Convolutional Neural Network. *Joint IEEE SCIS-ISIS*, Hachijo Island, Tokyo, Japan, 2020.
25. **Sakib, S.**, Siddique, M. A. B., & Rahman, M. A. Performance Evaluation of t-SNE and MDS Dimensionality Reduction Techniques with KNN, ENN and SVM Classifiers. *IEEE Region 10 Symposium (TENSYP 2020)*, Dhaka, Bangladesh, 2020.
26. Siddique, M. A. B., **Sakib, S.**, Khan, M. M. R., Tanzeem, A. K., Chowdhury, M., & Yasmin, N. Deep Convolutional Neural Networks Model-based Brain Tumor Detection in Brain MRI Images. *IEEE I-SMAC 2020*, Palladam, India, 2020.
27. Khan, M. M. R., **Sakib, S.**, Siddique, M. A. B., Chowdhury, M., Hossain, Z., Aziz, A., & Yasmin, N. Automatic Detection of COVID-19 Disease in Chest X-Ray Images using Deep Neural Networks. *IEEE R10 HTC 2020*, Kuching, Malaysia, 2020.
28. Siddique, M. A. B., **Sakib, S.**, & Rahman, M. A. Performance Analysis of Deep Autoencoder and NCA Dimensionality Reduction Techniques with KNN, ENN and SVM Classifiers. *IEEE International Conference on Innovation in Engineering and Technology (ICIET 2019)*, Dhaka, Bangladesh, 2019.
29. Siddique, F., **Sakib, S.**, & Siddique, M. A. B. Recognition of Handwritten Digit using Convolutional Neural Network in Python with Tensorflow and Comparison of Performance for Various Hidden Layers. *IEEE International Conference on Advances in Electrical Engineering (ICAEE 2019)*, Dhaka, Bangladesh, 2019.
30. Khan, M. M. R., **Sakib, S.**, Arif, R. B., & Siddique, M. A. B. Digital Image Restoration in Matlab: A Case Study on Inverse and Wiener Filtering. *International Conference on Innovation in Engineering and Technology (ICIET 2018)*, Dhaka, Bangladesh, 2018.

Selected Manuscripts Under Review

1. **Sakib, S.** & Roy, N. PhyResp: Robust Contactless Respiratory Sensing with Physics-Constrained Predictive State Modeling. *IEEE International Conference on Pervasive Computing and Communications (PerCom 2027)*.
2. **Sakib, S.** & Roy, N. Pixels-to-Phase: Reliability-Aware Latent State-Space Modeling for Contactless Respiratory Estimation. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027)*.
3. Sami, A. A., Chowdhury, M. J. U., Absar, S., Ahmed, I., **Sakib, S.**, Islam, M. R., Azam, S., & Roy, N. Vision-Language Models in Healthcare: A Comprehensive Review from Data to Diagnosis. *Artificial Intelligence in Medicine*.
4. **Sakib, S.** & Roy, N. SAGE-RR: Selective Auditing with Grounded Evidence for Robust Contactless Respiratory Rate

AWARDS, HONORS, AND COMPETITIVE SUPPORT

- **Doctoral Research Poster Award**, UMBC COEIT Research Day 2025, 2026
- **NSF Student Travel Grant**, IEEE/ACM CHASE 2026 2026
- **Student Travel Grant**, Information Systems Department, UMBC – IEEE ICMLA 2025 2025
- **Best Paper Award**, Signal Processing Track – IEEE IEMCON 2021
- **Best Paper Award**, Biomedical Engineering Track – IEEE R10 HTC 2021
- **Student Travel Grant**, University of Hyogo – IEEE SCIS & ISIS 2020 2020
- **HUMAP Fellowship**, University of Hyogo, Japan 2019–2020
- **Outstanding Chancellor's Teaching Assistant Award**, Department of EEE, IUBAT 2018–2020
- **University Merit Scholarship** – Undergraduate academic excellence (CGPA ≥ 3.9) 2016–2019
- **Dean's List of Honors** – all academic levels in undergraduate studies 2016–2019
- **Chancellor's Award for Academic Excellence** 2016
- **Departmental Chair's Award** – outstanding undergraduate academic performance 2015–2016

PEER REVIEW, PROGRAM COMMITTEES, AND PROFESSIONAL SERVICE

- **Journal Reviewer**: Applied Soft Computing; Array; Multimedia Tools and Applications; Biomedical Signal Processing and Control; Computational Biology and Chemistry; Computers in Biology and Medicine; Data in Brief; Frontiers in Computer Science; Healthcare Analytics; Informatics in Medicine Unlocked; Intelligence-Based Medicine; Pulmonary Medicine; Medical Engineering & Physics.
- **Conference and Workshop Reviewer**: ICVGIP (2026); IEEE BigData (2026); MICCAI (2026); AusDM (2026); NCA (2026); IEEE ICSIPA (2026); ICLR Workshop on TSALM (2026); CVPR Workshop on Med-Reasoner (2026); ICML Workshop on FMSD (2026); ICML Workshop on SD4H (2026); ICML Workshop on MusIML (2026); ECCV Workshop on HCV (2026); NeurIPS Workshop on Med-Reasoner (2026); NeurIPS Workshop on JUDGE (2026); NeurIPS Workshop on SocialAgent (2026); NeurIPS Workshop on MusIM (2025); IEEE SCIS-ISIS (2024, 2026).
- **Program Committee and Reviewer**, AAAI-26 Fall Symposium Series (FSS-26) 2026
- **Organizing Team Member**, NSF TRACE Testbed Workshop Spring 2026, UMBC Apr. 2026
- **Local Organizer**, IEEE PerCom 2025
- **Mentor**, IEEE Leading University Student Branch 2021–2022

TEACHING AND MENTORING EXPERIENCE

Graduate Teaching Assistant, Department of Information Systems Jan. 2024 – Present

University of Maryland, Baltimore County (UMBC) *Baltimore, Maryland*

- **IS 420: Database Application Development** – Aug. 2026–Present; Aug.–Dec. 2025; Jan.–May 2025. Led weekly lab sections for **80+ students** covering SQL and PL/SQL, ER modeling, normalization, stored procedures, triggers, and transaction control; designed and graded labs and examinations and evaluated course projects.
- **IS 620: Advanced Database Projects** – Jan.–May 2026; Aug.–Dec. 2024. Supported a project-based graduate course for **50+ students** in advanced database design, schema development, query optimization, and transaction processing; reviewed proposals and evaluated final deliverables.
- **IS 427: Introduction to Artificial Intelligence** – Jan.–May 2024. Assisted **45+ students** with search, probabilistic reasoning, supervised learning, and Python/scikit-learn programming projects.

Lecturer, Department of Computer Science and Engineering Oct. 2020 – Dec. 2022

Leading University *Sylhet, Bangladesh*

- Taught Machine Learning, Artificial Intelligence, Digital Signal Processing, Microprocessor and Interfacing, Electrical Circuits, Fundamentals of Computer Applications, and associated laboratory/project courses.
- Designed syllabi, labs, assignments, quizzes, midterms, final examinations, and project assessments; coordinated practical sessions and viva examinations.
- Supervised **13+ senior undergraduate theses**, resulting in multiple peer-reviewed publications; mentored students in experiment design, scientific writing, ethics, presentation, and camera-ready manuscript preparation.

TECHNICAL EXPERTISE

- **Programming and Scientific Computing:** Python, C/C++, MATLAB, SQL, Bash/Shell; NumPy, Pandas, SciPy, scikit-learn.
- **Deep Learning and Multimodal Foundation Models:** PyTorch, TensorFlow, Keras, Hugging Face Transformers; vision-language models (VLMs), multimodal large language models (MLLMs), video-time-series modeling, multimodal fusion, cross-modal attention, contrastive learning, diffusion models, and foundation-model adaptation.
- **Computer Vision and Medical Imaging:** OpenCV, torchvision, Albumentations, Detectron2; optical flow, motion and point tracking, object detection, segmentation, image registration, ROI extraction, video understanding, and medical/physiological image analysis.
- **Physiological Signal and Time-Series Analysis:** Contactless physiological sensing, respiratory-rate estimation, waveform reconstruction, temporal and spectral signal processing, signal-quality indices (SQI), multimodal sensor synchronization, temporal-frequency modeling, and physiological time-series learning.
- **Trustworthy, Reasoning, and Agentic AI:** Evidence-grounded multimodal reasoning, uncertainty and reliability estimation, calibration, selective auditing, counterfactual evaluation, active evidence acquisition, RAG, agentic AI workflows, and quantitative temporal reasoning.
- **Efficient and Edge AI:** NVIDIA Jetson Nano, Xavier, and Orin; CUDA/cuDNN; on-device inference, PEFT (LoRA/QLoRA), quantization, pruning, knowledge distillation, and resource-efficient model deployment.
- **Robotics and Cyber-Physical Sensing:** Boston Dynamics Spot, distributed/remote robotic experimentation, multimodal robotic sensing, camera-geometry and distance compensation, and edge-based autonomous physiological monitoring.
- **AI Systems and Reproducible Research:** Docker, Git/GitHub, MLflow, FastAPI, Linux; experiment design, ablation studies, cross-modal validation, model evaluation, and reproducible ML pipelines.

PROFESSIONAL MEMBERSHIPS

IEEE Graduate Student Member; ACM Student Member; ACM SIGIR Member; AAAI Student Member.